

TD2

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1. Fichier src/main.py de départ

```

@adlib1805-blip → /workspaces/esilv-marketpulse-g01-HANETeam (main) $ python src/main.py
cat src/main.py
    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

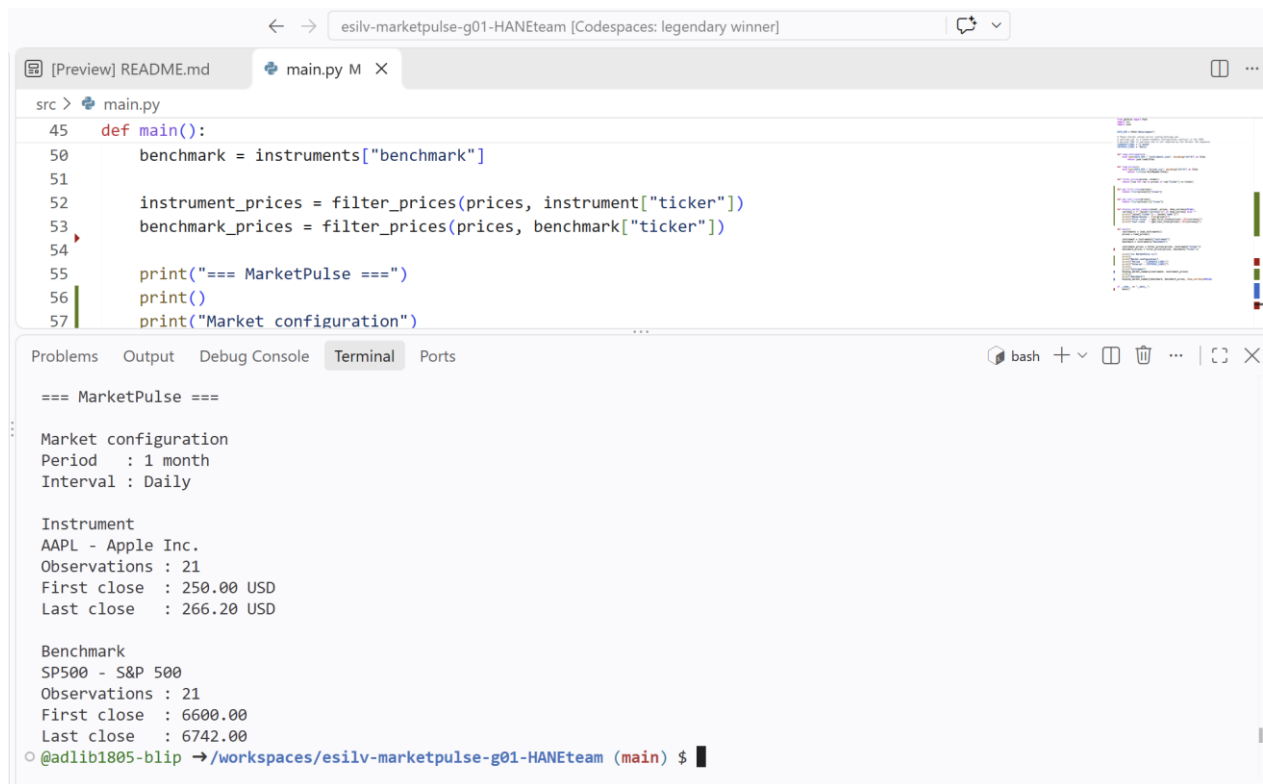
    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    instrument_latest = instrument_prices[-1]
    benchmark_latest = benchmark_prices[-1]

    print("=== MarketPulse ===")
    print()
    print("Instrument")
    print(f"{instrument['ticker']} - {instrument['name']}")
    print(f"Last price: {instrument_latest['close']} {instrument['currency']}")
    print()
    print("Benchmark")
    print(f"{benchmark['ticker']} - {benchmark['name']}")
    print(f"Last level: {benchmark_latest['close']}")
    print()
    print(f"Period: {LOOKBACK_LABEL}")
    print(f"Interval: {INTERVAL_LABEL}")
    print()
    print("Observations")
    print(f"{instrument['ticker']}: {len(instrument_prices)}")
    print(f"{benchmark['ticker']}: {len(benchmark_prices)}")

if __name__ == "__main__":
    main()
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```

2. Résultat du CORE : python src/main.py



The screenshot shows a CodeSpaces IDE window for a project named 'esilv-marketpulse-g01-HANETeam'. The top bar indicates the project name and a 'legendary winner' badge. Below the top bar, there are tabs for '[Preview] README.md' and 'main.py M X'. The 'main.py' tab is active, showing the following Python code:

```
src > main.py
45 def main():
50     benchmark = instruments["benchmark"]
51
52     instrument_prices = filter_prices(prices, instrument["ticker"])
53     benchmark_prices = filter_prices(prices, benchmark["ticker"])
54
55     print("=== MarketPulse ===")
56     print()
57     print("Market configuration")
```

Below the code editor, there is a 'Terminal' tab with the following output:

```
=== MarketPulse ===

Market configuration
Period : 1 month
Interval : Daily

Instrument
AAPL - Apple Inc.
Observations : 21
First close : 250.00 USD
Last close : 266.20 USD

Benchmark
SP500 - S&P 500
Observations : 21
First close : 6600.00
Last close : 6742.00
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```

The terminal output shows the results of the script execution, including market configuration and instrument/benchmark details.

3. Code final de src/main.py (CORE et optionnels)

```
from pathlib import Path
import csv
import json

DATA_DIR = Path("data/sample")

# These starter values mirror config/settings.yml.
# settings.yml is a human-readable configuration contract in the CORE.
# Parsing YAML is optional and is not required by the 18-hour lab sequence.
LOOKBACK_LABEL = "1 month"
INTERVAL_LABEL = "Daily"

def load_instruments():
    with open(DATA_DIR / "instruments.json", encoding="utf-8") as file:
        return json.load(file)

def load_prices():
    with open(DATA_DIR / "prices.csv", encoding="utf-8") as file:
        return list(csv.DictReader(file))

def filter_prices(prices, ticker):
    return [row for row in prices if row["ticker"] == ticker]

def get_first_close(prices):
    return float(prices[0]["close"])

def get_last_close(prices):
    return float(prices[-1]["close"])

def get_first_date(prices):
    return prices[0]["date"]

def get_last_date(prices):
    return prices[-1]["date"]

def get_min_close(prices):
    return min(float(row["close"]) for row in prices)

def get_max_close(prices):
    return max(float(row["close"]) for row in prices)

def count_positive_volume(prices):
    count = 0
    for row in prices:
        if int(row["volume"]) > 0:
            count += 1
    return count

def display_first_rows(prices, count=3):
    for row in prices[:count]:
        print(f"{row['date']} : {float(row['close']):.2f}")

def display_market_summary(asset, prices, show_currency=True):
```

```

currency = f" {asset['currency']}" if show_currency else ""
print(f"{asset['ticker']} - {asset['name']}")
print(f"Observations : {len(prices)}")
print(f"First date   : {get_first_date(prices)}")
print(f"Last date    : {get_last_date(prices)}")
print(f"First close   : {get_first_close(prices):.2f}{currency}")
print(f"Last close    : {get_last_close(prices):.2f}{currency}")
print(f"Min close     : {get_min_close(prices):.2f}{currency}")
print(f"Max close     : {get_max_close(prices):.2f}{currency}")

def main():
    instruments = load_instruments()
    prices = load_prices()

    instrument = instruments["instrument"]
    benchmark = instruments["benchmark"]

    instrument_prices = filter_prices(prices, instrument["ticker"])
    benchmark_prices = filter_prices(prices, benchmark["ticker"])

    print("=== MarketPulse ===")
    print()
    print("Market configuration")
    print(f"Period   : {LOOKBACK_LABEL}")
    print(f"Interval : {INTERVAL_LABEL}")
    print()
    print("Instrument")
    display_market_summary(instrument, instrument_prices)
    print(f"Positive volume days : {count_positive_volume(instrument_prices)}")
    print()
    print("First 3 closes")
    display_first_rows(instrument_prices)
    print()
    print("Benchmark")
    display_market_summary(benchmark, benchmark_prices, show_currency=False)

if __name__ == "__main__":
    main()

```

4. Résultat final avec les optionnels : python src/main.py

```
Problems  Output  Debug Console  Terminal  Ports  bash + - [ ] [X]

=== MarketPulse ===

Market configuration
Period   : 1 month
Interval : Daily

Instrument
AAPL - Apple Inc.
Observations : 21
First date   : 2026-09-01
Last date    : 2026-09-30
First close  : 250.00 USD
Last close   : 266.20 USD
Min close    : 249.80 USD
Max close    : 266.20 USD
Positive volume days : 21

First 3 closes
2026-09-01 : 250.00
2026-09-02 : 251.30
2026-09-03 : 249.80

Benchmark
SP500 - S&P 500
Observations : 21
First date   : 2026-09-01
Last date    : 2026-09-30
First close  : 6600.00
Last close   : 6742.00
Min close    : 6598.00
Max close    : 6742.00
@adlib1805-blip → /workspaces/esilv-marketpulse-g01-HANETeam (main) $
```